Md Ahsanul Haque, Ph.D.
MAT-120

Homework-2

Date: July 25, 2026

Name

Section

Instructions:

- Write your full name and section number inside the provided box.
- This assignment contains exactly 5 problems.
- Use correct mathematical notation and clearly show all logical steps.

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1. (10 points) Consider the function:

$$f(x) = \begin{cases} \dfrac{2x^3 - 5x}{x^3 + 1} & \text{if } x < 2 \\ 3 & \text{if } x = 2 \\ \dfrac{x^2 - x - 2}{x - 2} & \text{if } x > 2 \end{cases}$$

   (a) Find $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$

   (b) Does $\lim_{x \to 2} f(x)$ exist? Justify your answer.

   (c) Find $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to \infty} f(x)$

   (d) Identify all points of discontinuity and classify them.

2. (10 points)

   (a) Let $f(x) = \dfrac{\frac{1}{(x-1)^2} - \frac{1}{4}}{x + 1}$.
   Find $\lim_{x \to -1} f(x)$ using algebraic manipulation.

   (b) Determine if $g(x) = \dfrac{x^3 - 8}{x^2 - 4}$ is continuous at $x = 2$. If not, is the discontinuity removable?

   (c) Find all values of c that make $h(x) = \begin{cases} c^2 x + 2 & \text{if } x \leq 2 \\ cx^2 + 6 & \text{if } x > 2 \end{cases}$ continuous everywhere.

   (d) Investigate the solution of $x^4 - 4x^3 + 2 = 0$ in the interval $[0, 1]$.

3. (10 points) Analyze the following limits:
   (a) $\lim_{x\to 0} \dfrac{1-\cos(5x)}{3x^2}$
   (b) $\lim_{x\to -3} \dfrac{x+3}{\sqrt{x^2+7}-4}$
   (c) $\lim_{x\to -\infty} \frac{\sqrt{25x^2+3x}}{4x-1}$
   (d) $\lim_{x\to 0} x^4 \sin\left(\frac{\pi}{x}\right)$ using the Squeeze Theorem.

4. (10 points) Let $f(x) = x^3 - 6x^2 + 9x$ and $g(x) = \sqrt{3x+4}$
   (a) Find the equation of the tangent line of f at $x = 1$.
   (b) Find the average rate of change of f on $[0, 2]$ and compare it with the instantaneous rate at $x = 1$.
   (c) Use the **limit definition** of the derivative to find $g'(x)$. Show the rationalization step clearly.
   (d) Find the equation of the normal line (the line perpendicular to the tangent line) of g at $x = 4$.

5. (10 points) Consider the curve defined by
$$3x^2 - 2xy + 2y^2 = 14.$$
   (a) Find $\dfrac{dy}{dx}$ using implicit differentiation.
   (b) Find all points where the tangent line is horizontal.
   (c) Find all points where the tangent line is vertical.
   (d) Find $\dfrac{d^2y}{dx^2}$ in terms of x and y.